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CSD024

TripConnect

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CONTENTS

INTRODUCTION	3
PURPOSE AND SCOPE	4
PRODUCT FUNCTION	5
USE CASE DIAGRAM	6
HARDWARE REQUIREMENTS	7
SOFTWARE REQUIREMENTS	8
TECH STACK	9
DIAGRAMS • DATA FLOW DIAGRAM	10
• ACTIVITY DIAGRAM	11
• SEQUENCE DIAGRAM	14
CLASS DIAGRAM	19
DATABASE	20
SCREEN IMAGE	27
REFERENCE	29

INTRODUCTION

TripConnect is a smart travel platform that makes planning journey simple and exciting .Users can design their own trips by choosing destinations, dates, and budgets, or explore ready-made packages for solo, family, or group travel. The platform also connects like-minded traveler's, making journeys more fun and social. With secure payments, easy bookings, and helpful notifications, TripConnect is your all-in-one travel companion.

PURPOSE

• Simplify the travel planning process for users through a centralized platform that supports both customized trip creation and inbuilt package selection.

• Enable travellers to create profiles, set preferences, and connect with compatible travel companions for shared journeys.

SCOPE

• Let users create profiles with travel preferences.

• Match users with suitable travel buddies.

• Enable customized trip planning and provide predefined packages.

• Support trip booking, secure payments, and notifications.

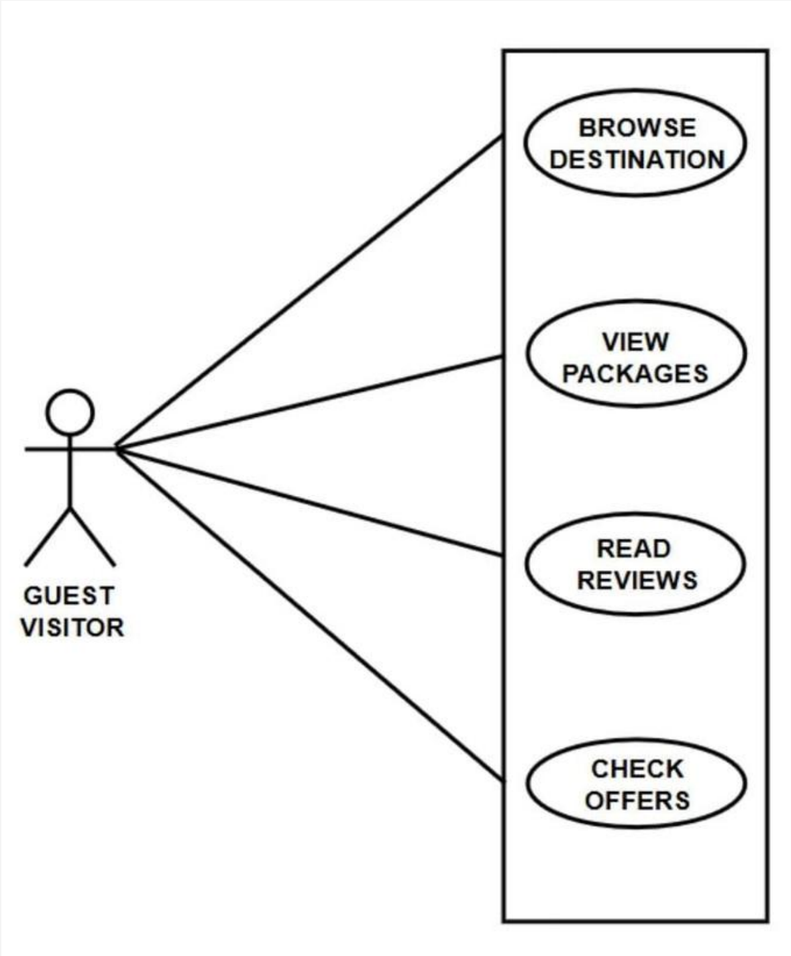
• Allow communication between travellers and include reviews/ratings.

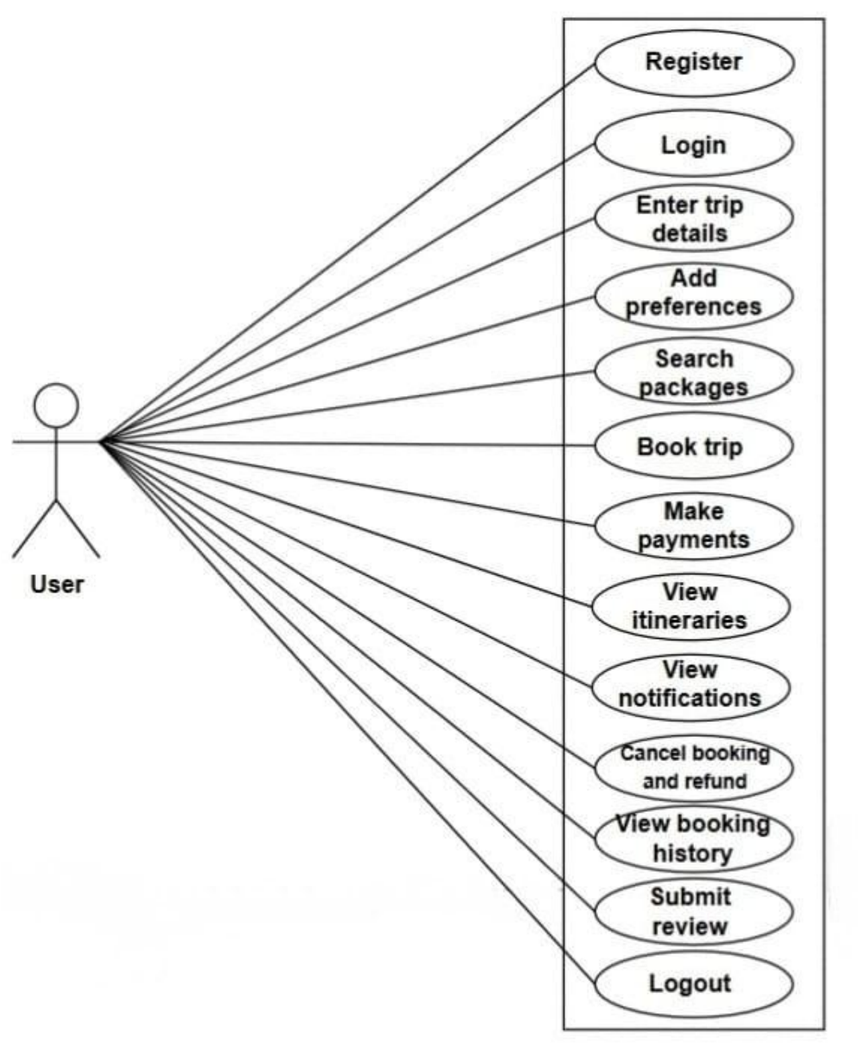
PRODUCT FUNCTION

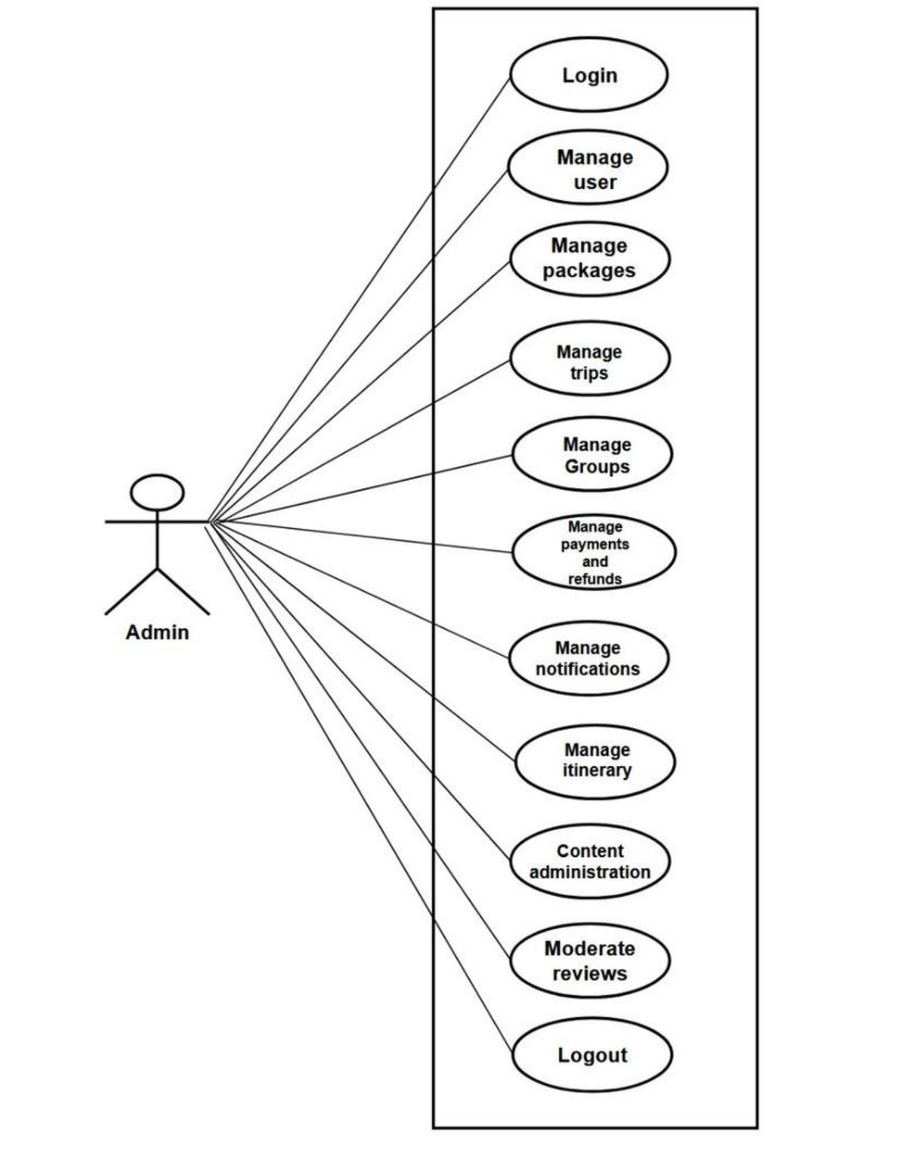
TripConnect is a travel platform that makes trip planning, booking, and networking easy for travelers.

- **User Profiles** – Create and manage accounts with travel preferences and past trips.
- **Package Booking** – Browse, view details, and book ready-made packages.
- **Custom Trips** – Plan personalized itineraries based on budget and schedule.
- **Matchmaking & Search** – Find and connect with travelers with similar interests and filter packages by need.
- **Notifications** – Get alerts for bookings, updates, and messages.
- **Admin Tools** – Manage users, trips, and packages while ensuring platform security.
- **Payments** – Support secure online payments for bookings and trips.

USE CASE DIAGRAMS







HARDWARE INTERFACES

TripConnect is a software platform accessed via standard computing and mobile devices. The system does not require any specialized hardware. The hardware interface considerations are as follows:

- **User Devices**

Supported devices include desktops, laptops, tablets, and smartphones.

Interfaces are standard input/output devices: touchscreen, keyboard, mouse.

- **Networking Hardware**

Requires standard network connectivity (Wi-Fi, 4G/5G, Ethernet) to access the platform.

Communication occurs over standard web protocols (HTTP/HTTPS).

- **Payment Processing**

Interacts with online payment gateways via the internet.

No direct connection to physical card readers or other specialized payment hardware is needed.

Conclusion

TripConnect has no dedicated hardware interface requirements beyond standard computing devices and network connectivity.

SOFTWARE INTERFACES

- **MongoDB** – Serves as the primary database for storing user details, packages, groups, payments, chats, and travel history. Communication is handled via the MongoDB Node.js driver with data exchanged in BSON/JSON format.

- **Stripe API** – Provides secure online payment processing for bookings, cancellations, and refunds. Interaction occurs through REST APIs over HTTPS, with requests and responses formatted in JSON, ensuring safe and reliable transactions.

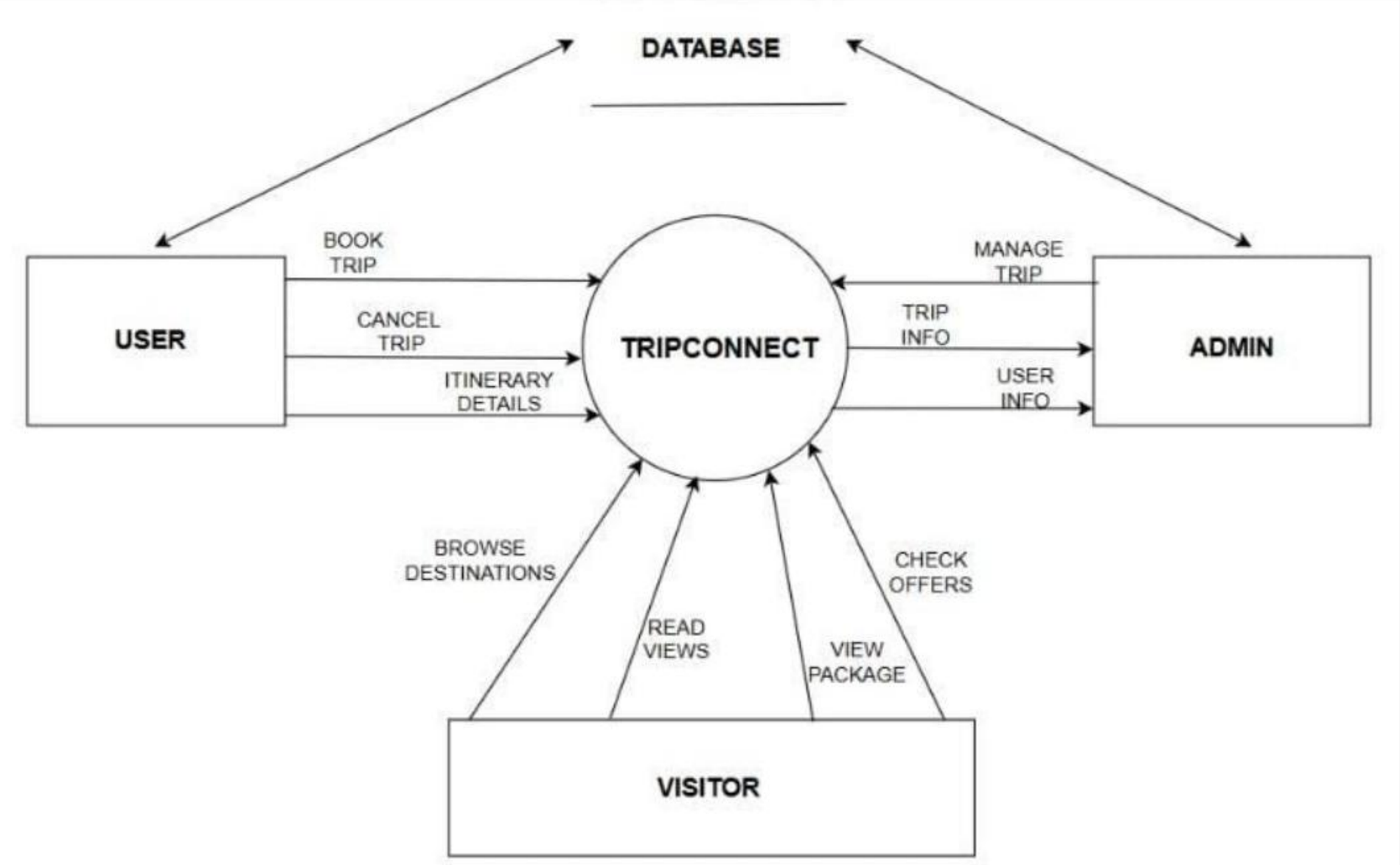
- **Google Maps API** – Enables TripConnect to display destinations, routes, and itineraries. The system communicates with the API using HTTPS requests and receives JSON-formatted responses to visualize trip details.

TECH STACK

The TripConnect website will be built using the following key technologies:

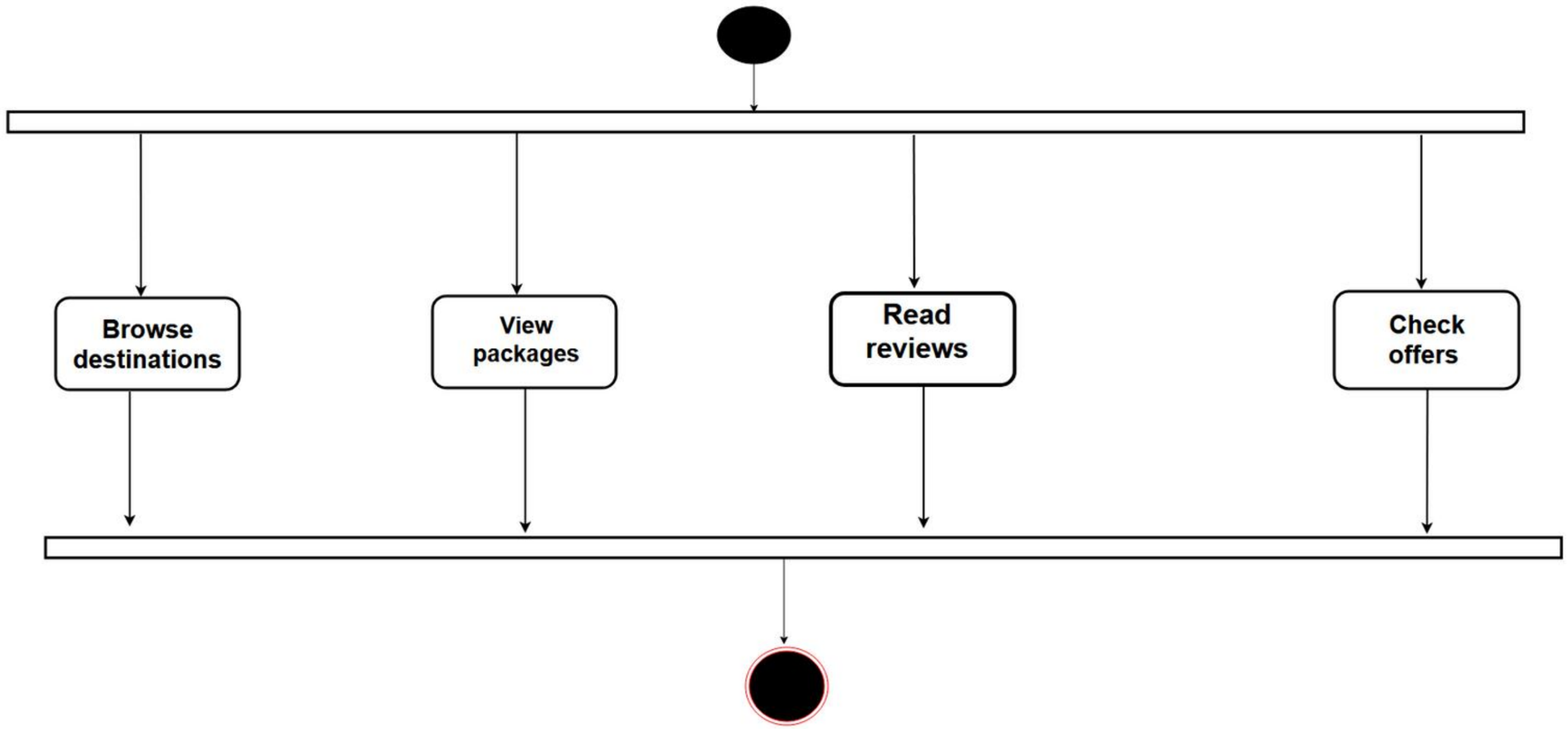
- FRONTEND TOOLS- React.js / Redux / Material UI / Axios.
- BACKEND TOOLS- Node.js / Express.js / Mongoose / MongoDB.
- OTHER TOOLS AND TECHNOLOGIES- Stripe API / Twilio / Socket IO / Navigation / AI-ML

DFD - LEVEL 0

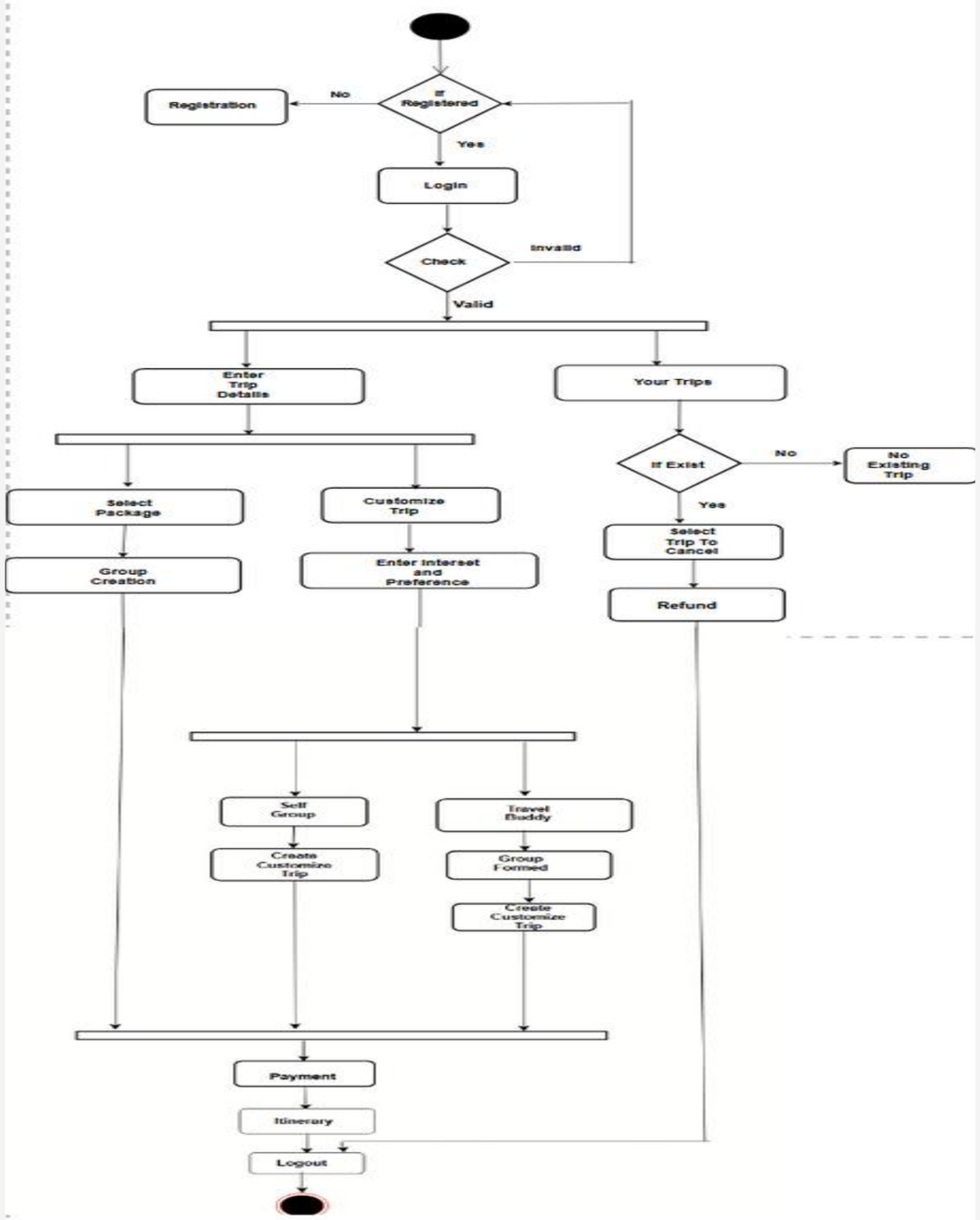

DATABASE
USER
BOOK TRIP
CANCEL TRIP
ITINERARY DETAILS
TRIPCONNECT
MANAGE TRIP
TRIP INFO
USER INFO
ADMIN
BROWSE DESTINATIONS
READ VIEWS
VIEW PACKAGE
CHECK OFFERS
VISITOR

ACTIVITY

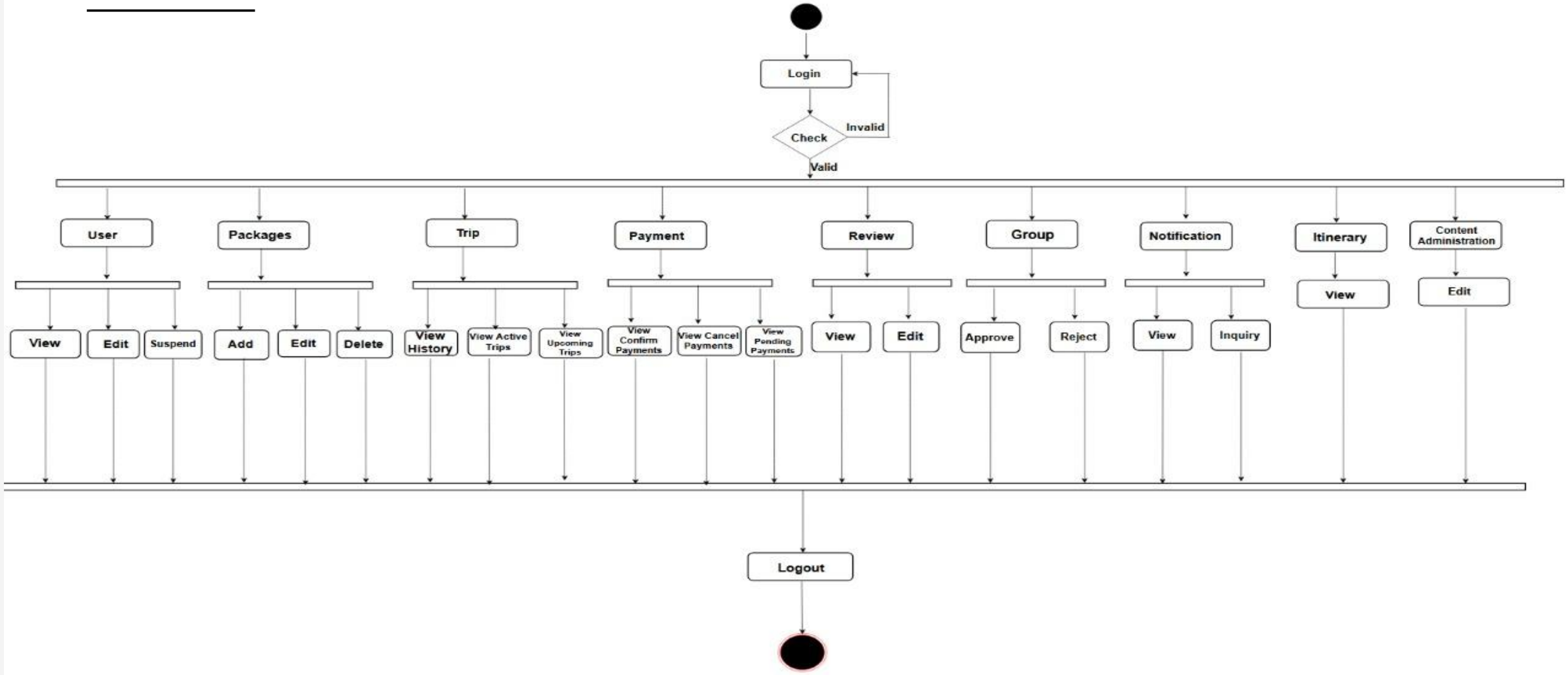
GUEST VISITOR



USER

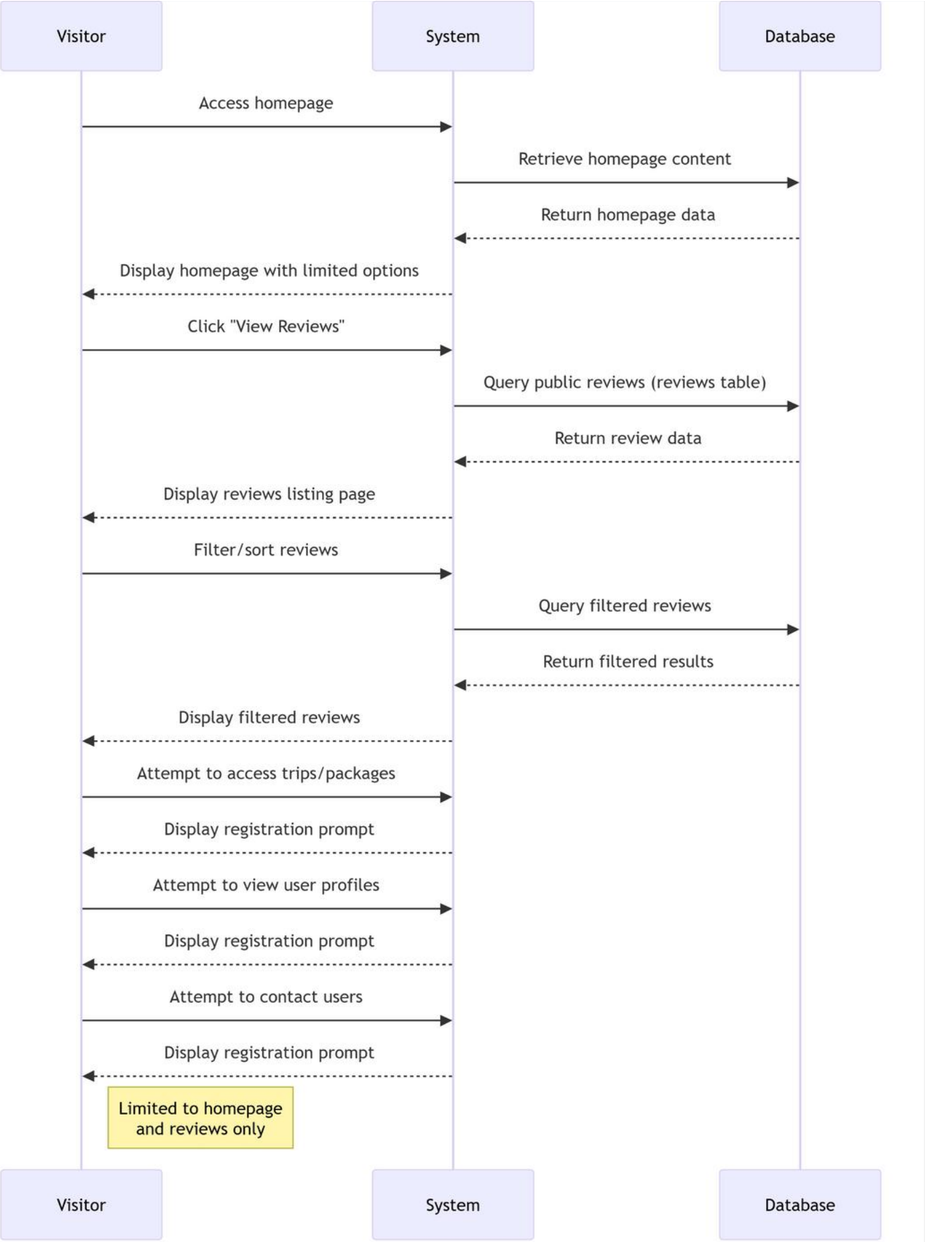

Registration
No
If Registered
Yes
Login
Check
Invalid
Valid
Enter Trip Details
Your Trips
If Exist
No
No Existing Trip
Yes
Select Trip To Cancel
Refund
Select Package
Customize Trip
Group Creation
Enter Interest and Preference
Self Group
Travel Buddy
Create Customize Trip
Group Formed
Create Customize Trip
Payment
Itinerary
Logout

ADMIN

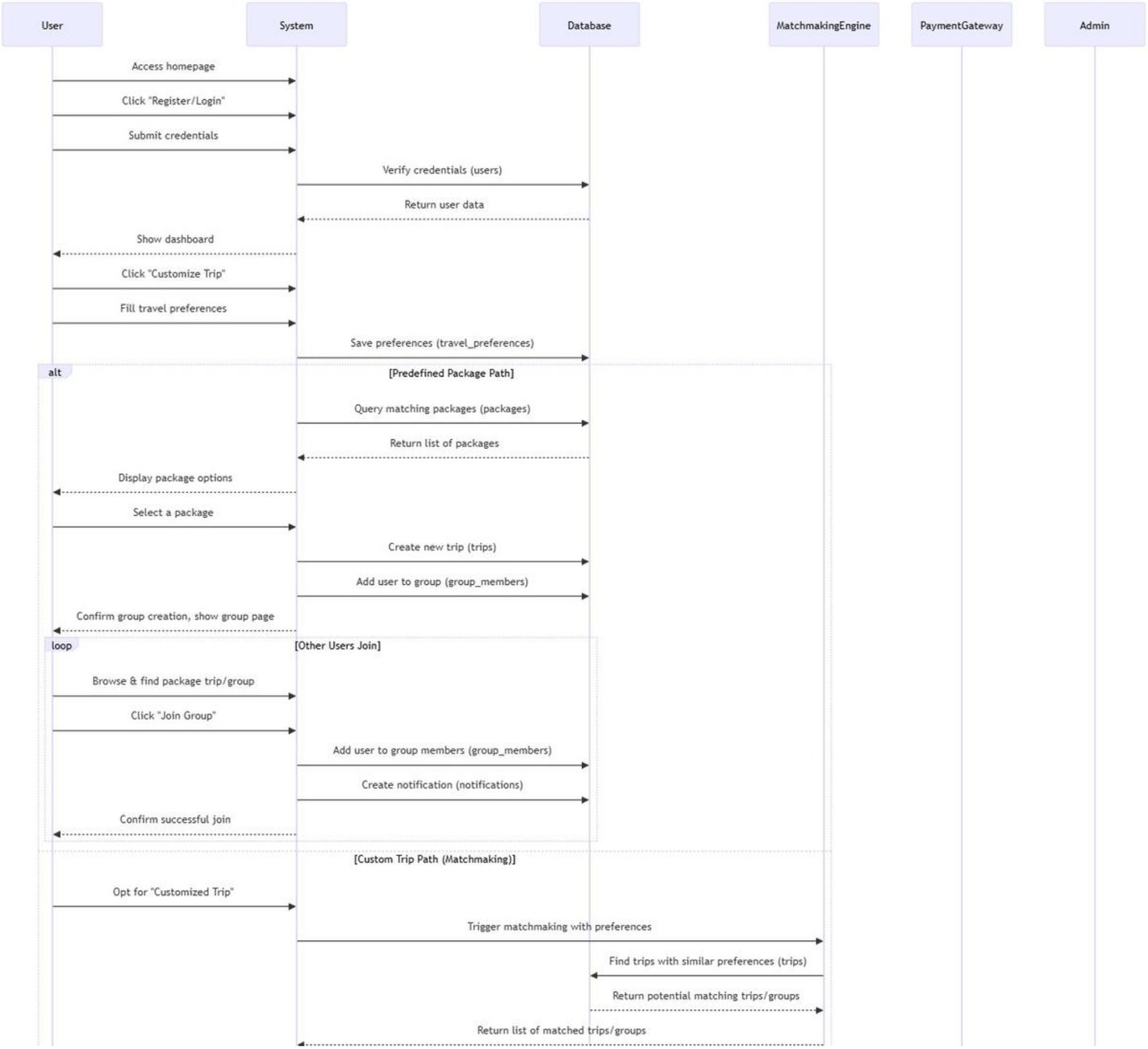

Login
Check
Invalid
Valid
User
Packages
Trip
Payment
Review
Group
Notification
Itinerary
Content Administration
View
Edit
Suspend
Add
Edit
Delete
View History
View Active Trips
View Upcoming Trips
View Confirm Payments
View Cancel Payments
View Pending Payments
View
Edit
Approve
Reject
View
Inquiry
View
Edit
Logout

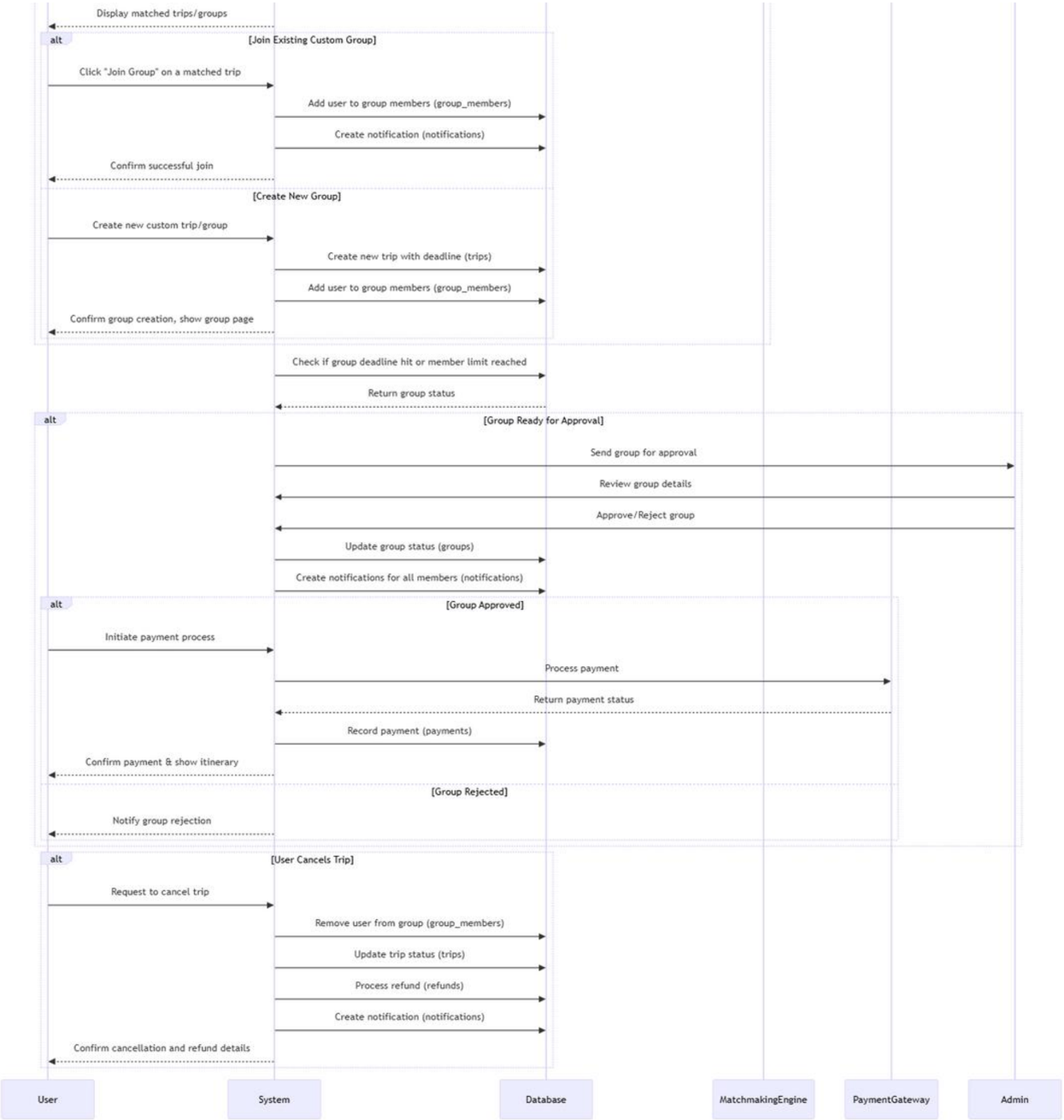
SEQUENCE DIAGRAMS

Visitor

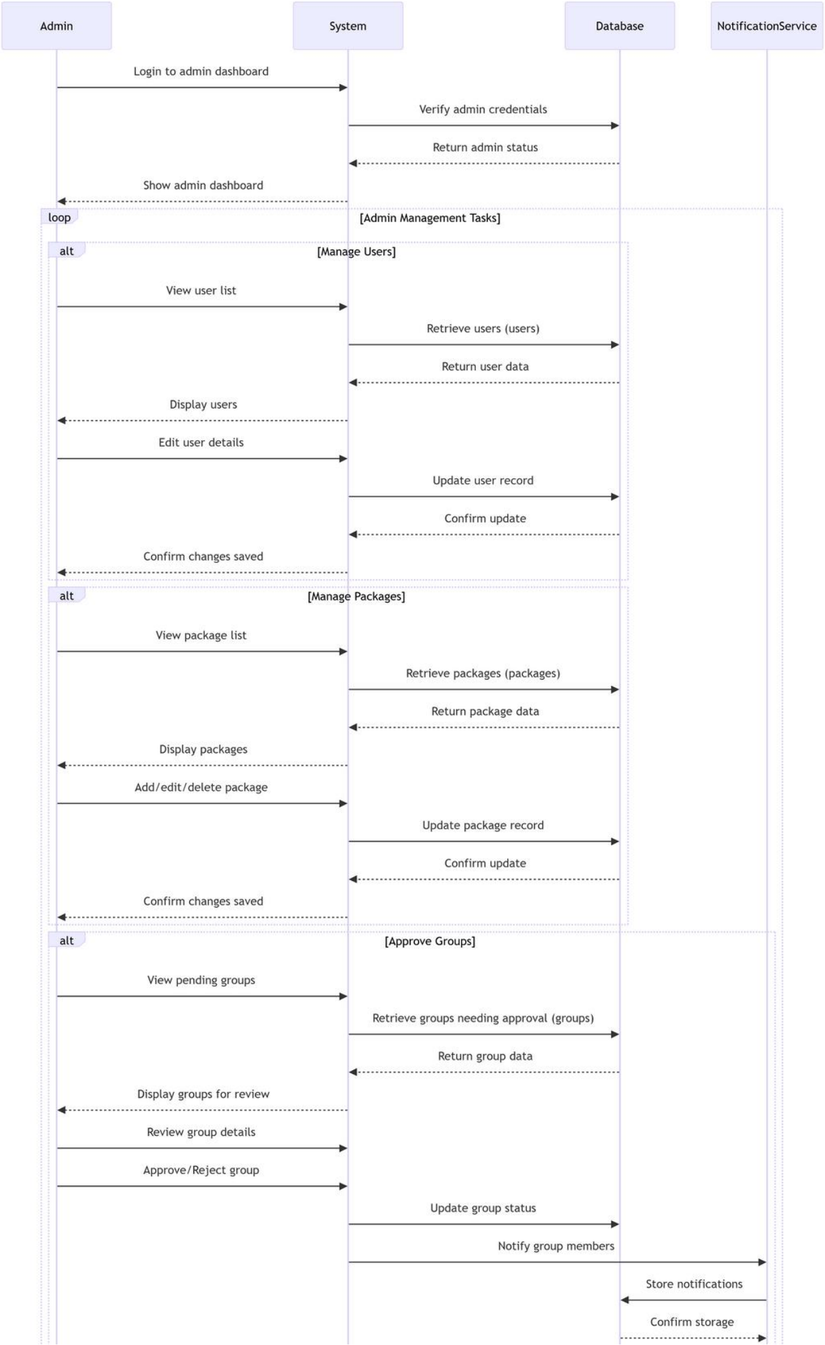


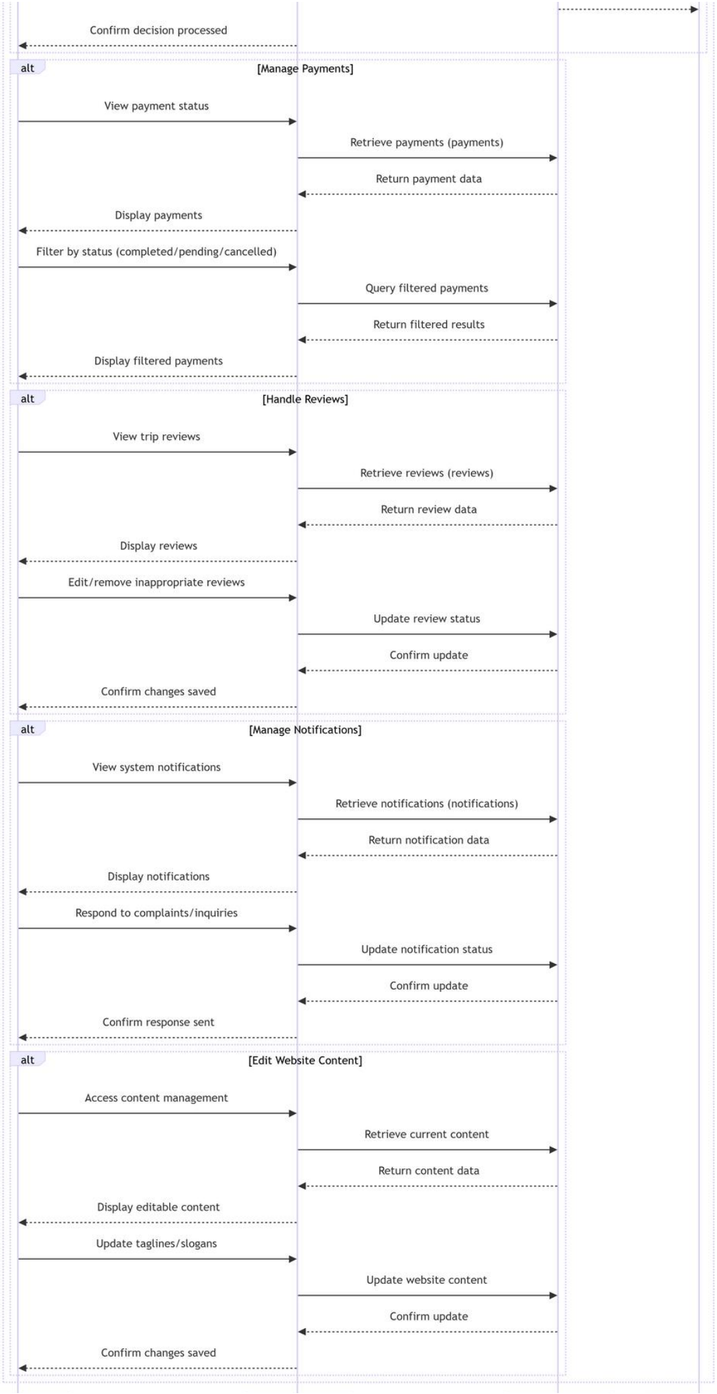
User


User
System
Database
MatchmakingEngine
PaymentGateway
Admin
Access homepage
Click "Register/Login"
Submit credentials
Verify credentials (users)
Return user data
Show dashboard
Click "Customize Trip"
Fill travel preferences
Save preferences (travel_preferences)
alt
[Predefined Package Path]
Query matching packages (packages)
Return list of packages
Display package options
Select a package
Create new trip (trips)
Add user to group (group_members)
Confirm group creation, show group page
loop
[Other Users Join]
Browse & find package trip/group
Click "Join Group"
Add user to group members (group_members)
Create notification (notifications)
Confirm successful join
[Custom Trip Path (Matchmaking)]
Opt for "Customized Trip"
Trigger matchmaking with preferences
Find trips with similar preferences (trips)
Return potential matching trips/groups
Return list of matched trips/groups

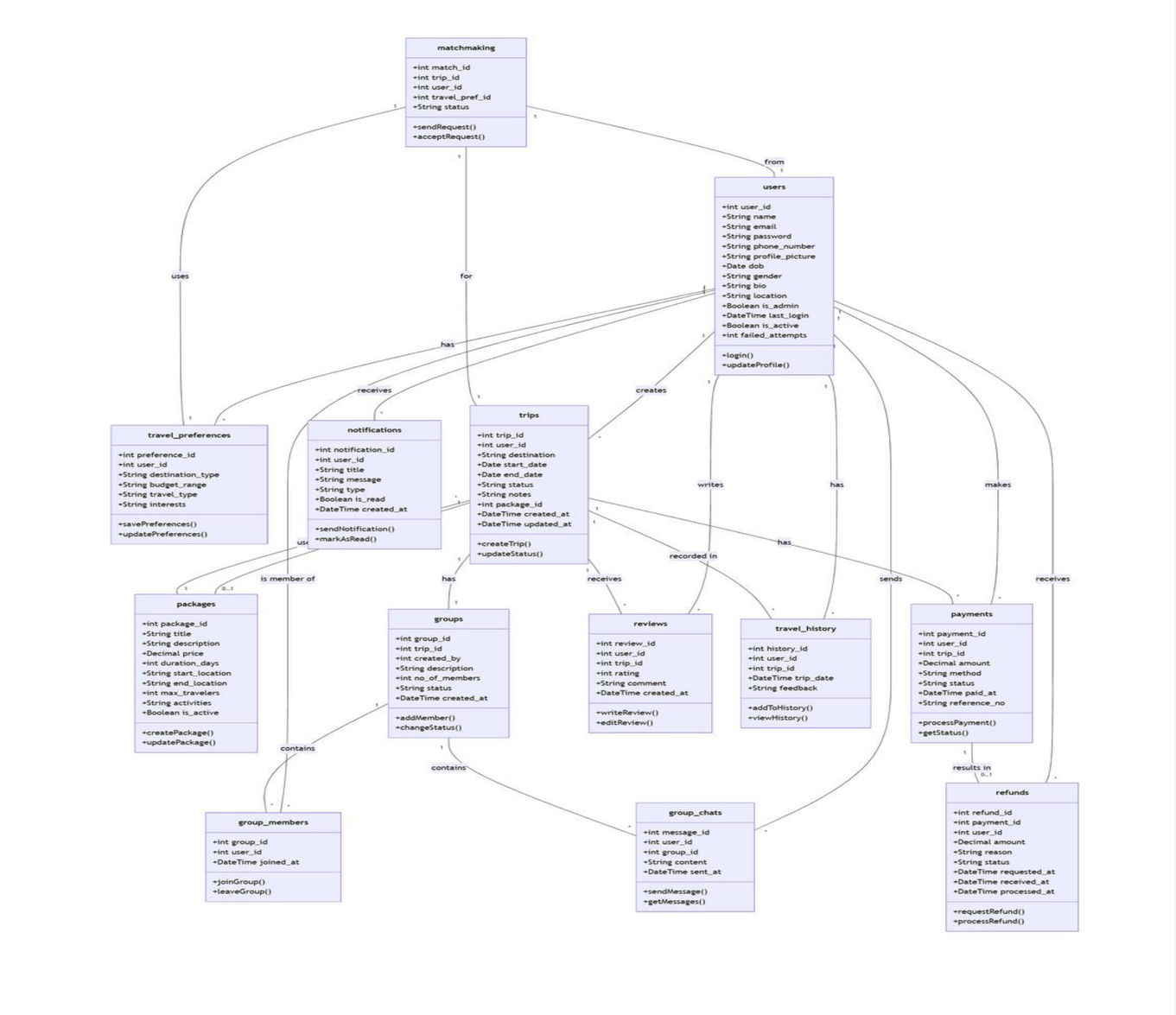

Display matched trips/groups
alt
[Join Existing Custom Group]
Click "Join Group" on a matched trip
Add user to group members (group_members)
Create notification (notifications)
Confirm successful join
[Create New Group]
Create new custom trip/group
Create new trip with deadline (trips)
Add user to group members (group_members)
Confirm group creation, show group page
Check if group deadline hit or member limit reached
Return group status
alt
[Group Ready for Approval]
Send group for approval
Review group details
Approve/Reject group
Update group status (groups)
Create notifications for all members (notifications)
alt
[Group Approved]
Initiate payment process
Process payment
Return payment status
Record payment (payments)
Confirm payment & show itinerary
[Group Rejected]
Notify group rejection
alt
[User Cancels Trip]
Request to cancel trip
Remove user from group (group_members)
Update trip status (trips)
Process refund (refunds)
Create notification (notifications)
Confirm cancellation and refund details
User
System
Database
MatchmakingEngine
PaymentGateway
Admin

Admin


Admin
System
Database
NotificationService
Login to admin dashboard
Verify admin credentials
Return admin status
Show admin dashboard
loop
[Admin Management Tasks]
alt
[Manage Users]
View user list
Retrieve users (users)
Return user data
Display users
Edit user details
Update user record
Confirm update
Confirm changes saved
alt
[Manage Packages]
View package list
Retrieve packages (packages)
Return package data
Display packages
Add/edit/delete package
Update package record
Confirm update
Confirm changes saved
alt
[Approve Groups]
View pending groups
Retrieve groups needing approval (groups)
Return group data
Display groups for review
Review group details
Approve/Reject group
Update group status
Notify group members
Store notifications
Confirm storage


Confirm decision processed
alt
[Manage Payments]
View payment status
Retrieve payments (payments)
Return payment data
Display payments
Filter by status (completed/pending/cancelled)
Query filtered payments
Return filtered results
Display filtered payments
alt
[Handle Reviews]
View trip reviews
Retrieve reviews (reviews)
Return review data
Display reviews
Edit/remove inappropriate reviews
Update review status
Confirm update
Confirm changes saved
alt
[Manage Notifications]
View system notifications
Retrieve notifications (notifications)
Return notification data
Display notifications
Respond to complaints/inquiries
Update notification status
Confirm update
Confirm response sent
alt
[Edit Website Content]
Access content management
Retrieve current content
Return content data
Display editable content
Update taglines/slogans
Update website content
Confirm update
Confirm changes saved

Class Diagram



DATABASE TABLES

USER TABLE

Field	Type	Description
user_id	Number (Auto Increment)	Primary key
name	String	User's full name
email	String (Unique)	User's email
phone number	String (Unique)	User's phone number
password	String	Hashed password
profile picture	String	Path to profile image
dob	Date	Date of birth
gender	Enum('Male','Female',' Other')	Gender
bio	String	Biography
location	String	Location
stamin	Boolean	Admin status
last login	DateTime	Last login timestamp
is_active	Boolean	Active status
failed_attempts	Number	Failed login attempts

TRIPS

Field	Type	Description
preference_id	Number (Auto Increment)	Primary Key
user_id	Number (Ref: Users)	User Reference
destination_type	String	Preferred destination type
budget_range	String	Budget range
travel_type	Enum ('Solo','Group','Famil y')	Type of travel
interests	String	User's travel interests

DATABASE TABLES

TRAVEL PREFERENCE

Field	***Type***	***Description***
preference_id	*Number (Auto Increment)*	*Primary Key*
user_id	*Number (Ref: Users)*	*User Reference*
destination_type	*String*	*Preferred destination type*
budget_range	*String*	*Budget range*
travel_type	*Enum ('Solo','Group','Family')*	*Type of travel*
interests	*String*	*User's travel interests*

GROUPS

Field	***Type***	***Description***
group_id	*Number (Auto Increment)*	*Primary key*
trip_id	*Number (Ref:trips)*	*Trip reference*
created_by	*Number (Ref:users)*	*Creator user interface*
description	*String*	*Group description*
no_of_members	*Number*	*Number of members*
created_at	*DateTime*	*Creation timestamp*

DATABASE TABLES

GROUP CHATS

Field	***Type***	***Description***
group_id	*Number (Auto Increment)*	*Primary key*
trip_id	*Number (Ref:trips)*	*Trip reference*
created_by	*Number (Ref:users)*	*Creator user interface*
description	*String*	*Group description*
no_of_members	*Number*	*Number of members*
created_at	*DateTime*	*Creation timestamp*

MATCHMAKING

Field	*Type*	*Description*
match_id	*Number (Auto Increment)*	*Primary key*
trip_id	*Number(Ref:trips)*	*Trip reference*
user_id	*Number(Ref:users)*	*User reference*
travel_pref_id	*Number(Ref:travel_ preferences*	*Travel preference reference*

DATABASE TABLES

PACKAGES

Field	*Type*	*Description*
match_id	*Number (Auto Increment)*	*Primary key*
trip_id	*Number(Ref:trips)*	*Trip reference*
user_id	*Number(Ref:users)*	*User reference*
travel_pref_id	*Number(Ref:travel _preferences*	*Travel preference reference*

GROUP CHATS

Field	*Type*	*Description*
message_id	*Number(Auto Increment)*	*Primary key*
group_id	*Number (Ref:groups)*	*Group reference*
sender_id	*Number(Ref:users)*	*Sender reference*
content	*String*	*Chat message content*
sent_at	*DateTime*	*Message timestamp*

DATABASE TABLES

PAYMENTS

Field	*Type*	*Description*
payment_id	*Number (Auto Increment)*	*Primary key*
user_id	*Number (Ref:users)*	*User reference*
trip_id	*Number(Ref:trips)*	*Trip reference*
amount	*Decimal*	*Payment amount*
method	*String*	*Payment method*
status	*Enum('Pending,' Completed','Failed')*	*Payment status*
paid_at	*DateTime*	*Payment timestamp*
reference_no	*String*	*Payment reference number*

REFUNDS

Field	*Type*	*Description*
payment_id	*Number (Auto Increment)*	*Primary key*
user_id	*Number (Ref:users)*	*User reference*
trip_id	*Number(Ref:trips)*	*Trip reference*
amount	*Decimal*	*Payment amount*
method	*String*	*Payment method*
status	*Enum('Pending,'Completed','Failed')*	*Payment status*
paid_at	*DateTime*	*Payment timestamp*
reference_no	*String*	*Payment reference number*

DATABASE TABLES

REVIEWS

Field	*Type*	*Description*
payment_id	*Number (Auto Increment)*	*Primary key*
user_id	*Number (Ref:users)*	*User reference*
trip_id	*Number(Ref:trips)*	*Trip reference*
amount	*Decimal*	*Payment amount*
method	*String*	*Payment method*
status	*Enum('Pending,'Completed','Failed')*	*Payment status*
paid_at	*DateTime*	*Payment timestamp*
reference_no	*String*	*Payment reference number*

TRAVEL HISTORY

Field	*Type*	*Description*
history_id	*Number (Auto Increment)*	*Primary key*
user_id	*Number (Ref:users)*	*User reference*
trip_id	*Number (Ref:trips)*	*Trip reference*
trip_date	*DateTime*	*Date of trip*
feedback	*String*	*User feedback*

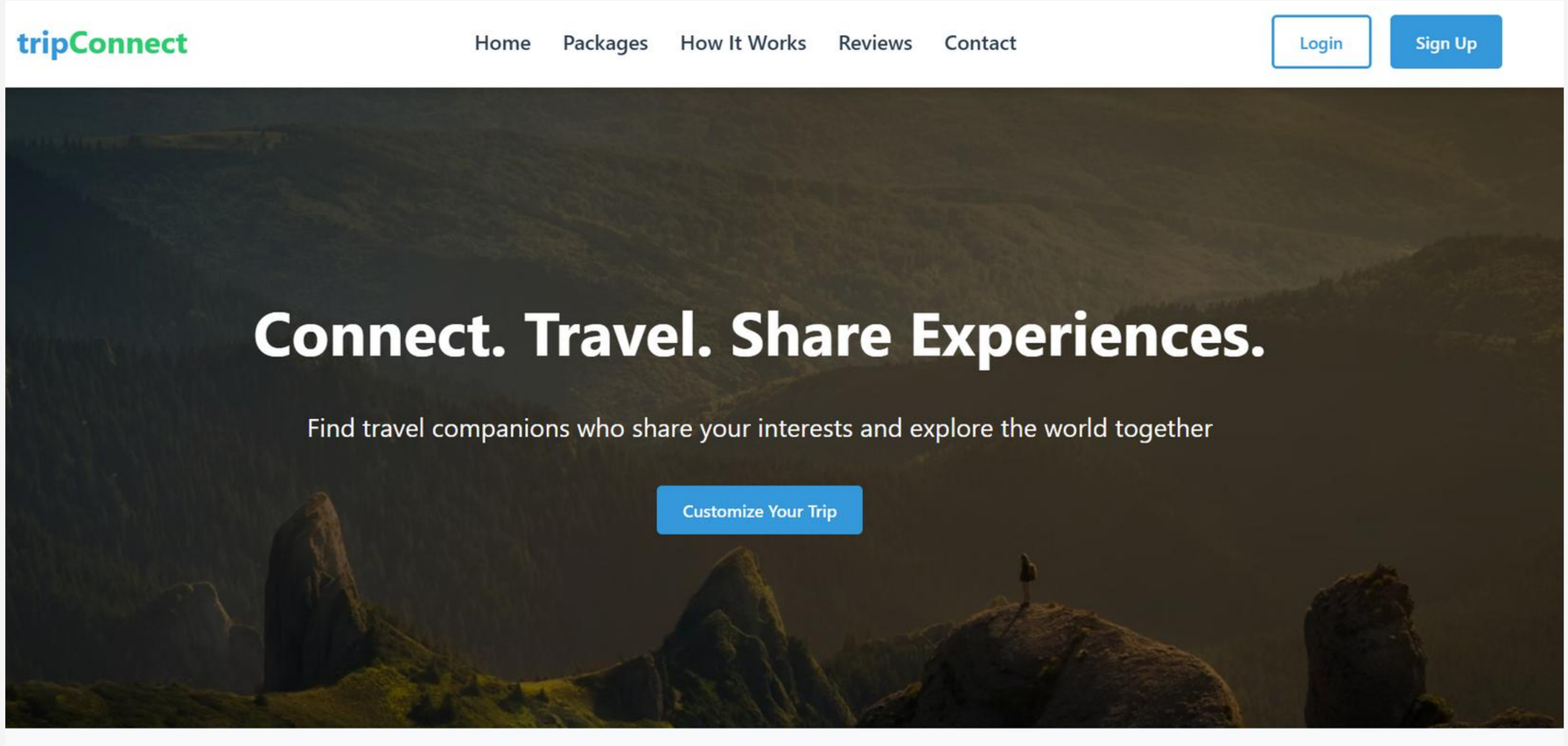
DATABASE TABLES

NOTIFICATION

Field	*Type*	*Description*
notification_id	*Number (Auto Increment)*	*Primary key*
user_id	*Number (Ref:users)*	*User reference*
title	*String*	*Notification title*
message	*String*	*Notification message*
type	*Enum('trip','review','message','admin_alert')*	*Notification type*
is_read	*Boolean*	*Read status*
created_at	*DateTime*	*Created timestamp*

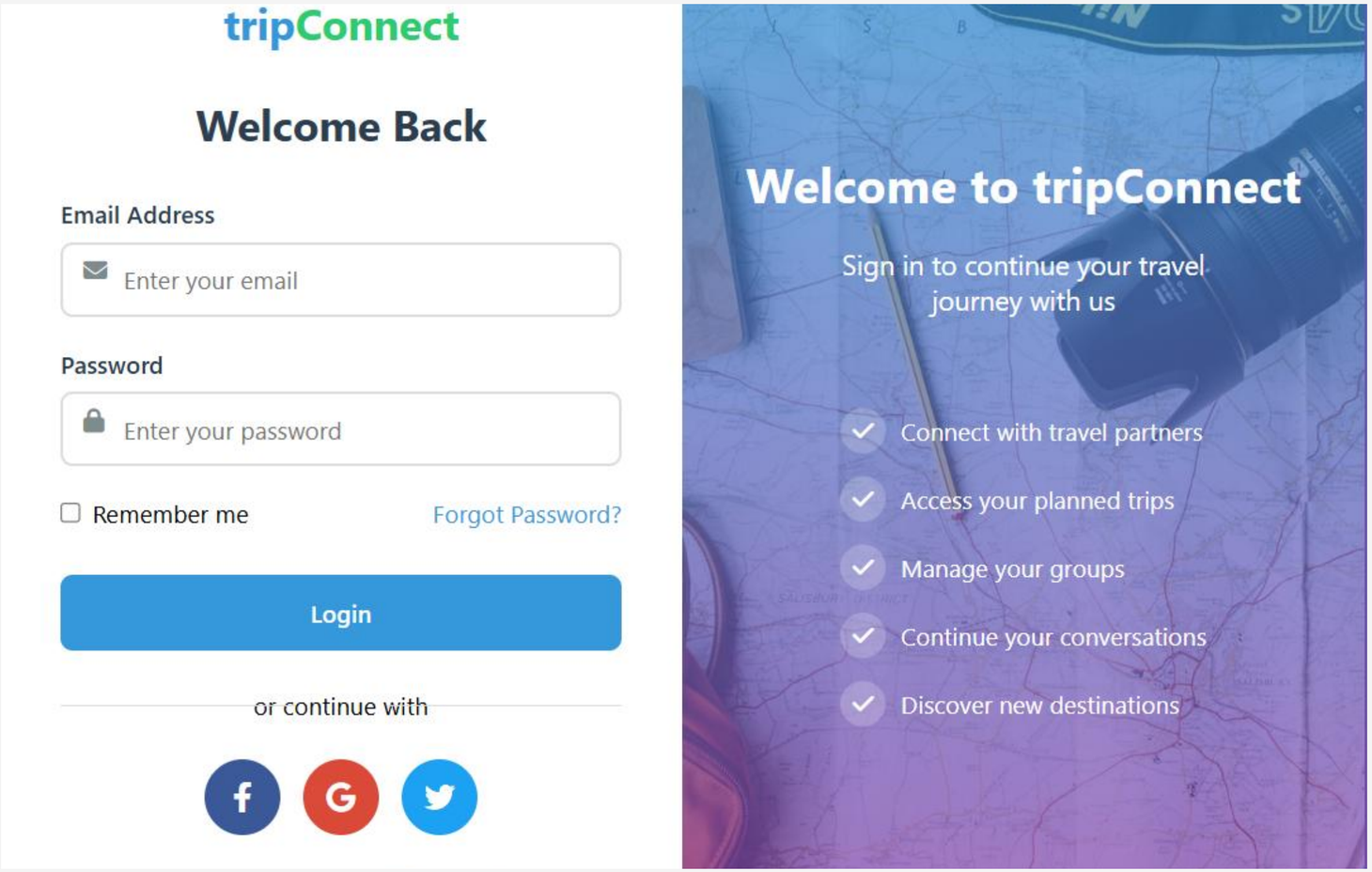
SCREEN IMAGES

Homepage



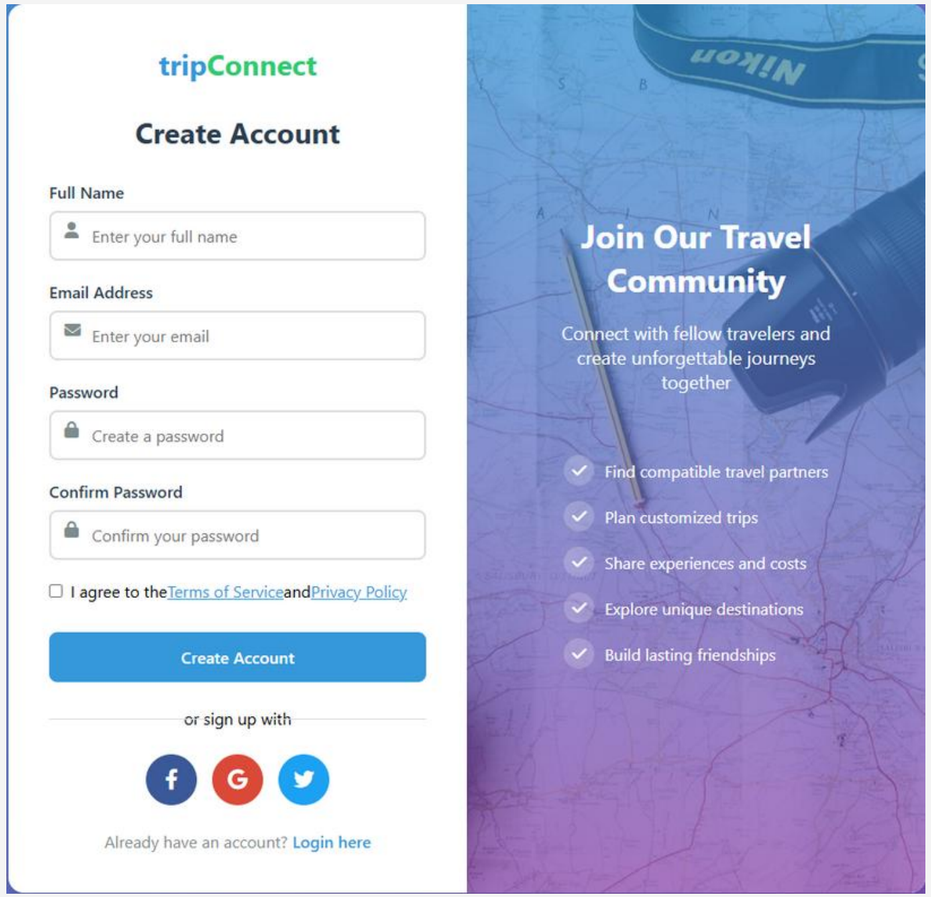
SCREEN IMAGES

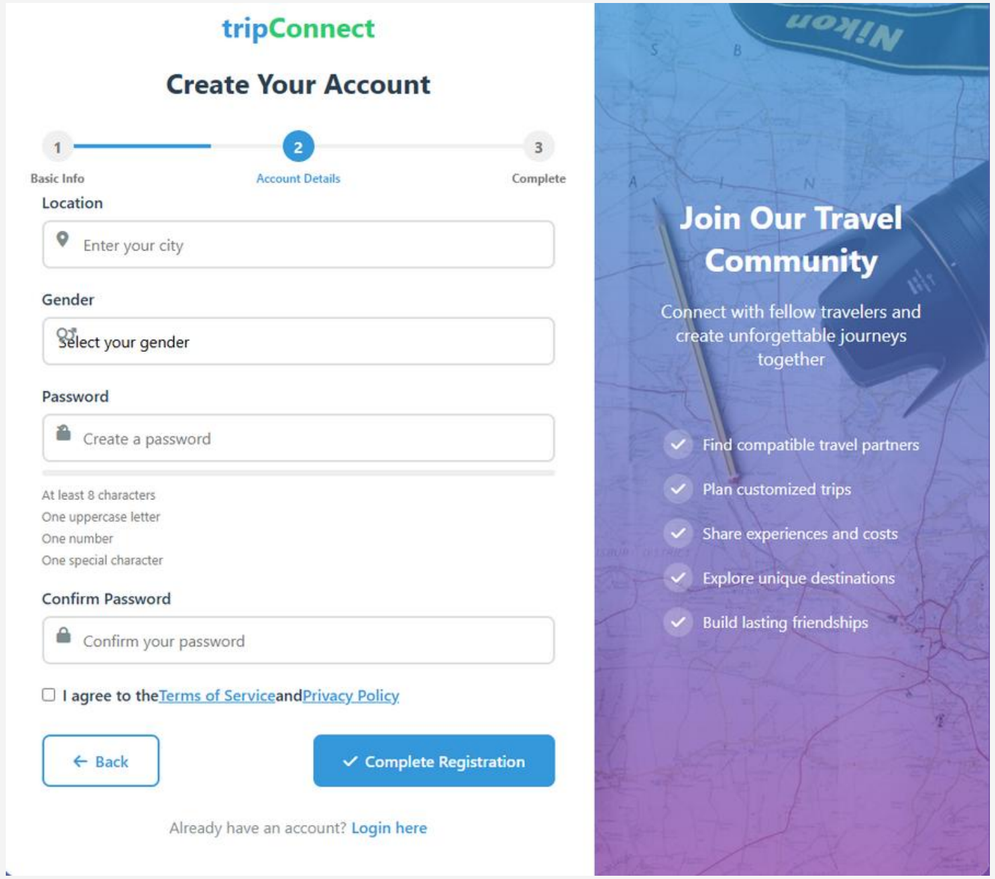
Login



SCREEN IMAGES

Register





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THANK YOU